

Computational Supplement: Correlation Emergence and the Epps Effect

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v2.2.0 — Numerical resolution update
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Abstract

This supplement records the analytical and simulation companion to “Correlation emergence and the Epps effect in two coupled limit order books” (arXiv:2606.14182). It evaluates the paper’s ordinary and fractional correlation build-up factors, reports their sensitivities and identifiability limits, diagnoses finite-grid boundary representation, and verifies a uniform-operational simulation followed by explicit book-specific previous-refresh subordination. The numerical evidence includes clock-only and translation-mode coupling comparisons, a combined fixed-parameter comparison, single-trade and meta-order impact, mid-price-increment and trade-sign autocorrelations, fixed-time order-book shock recovery, and stylised-facts diagnostics. The observation-clock comparison separates exogenous long-memory order flow from Poisson, Mittag-Leffler and exponentially tempered observation clocks, then measures paired single-trade and meta-order impact in each domain. This numerical resolution update increases independent simulation depth, refines the impact grid, and reports group-aware mean uncertainty. The computations are model-conditional reproducibility evidence, not empirical calibration or mathematical proof.

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1 Scientific scope

The frozen source paper remains the mathematical source of truth. The simulation implementation preserves that source snapshot. The ordinary clock, coupling, and combined curves remain illustrations of the paper’s renewal-envelope, exponential-response, and leading-order separability assumptions. The fractional curves evaluate the stated Mittag–Leffler replacement on the non-negative real axis. The reaction-boundary implementation uses the decaying Gaussian convention for which the paper gives a finite half-line moment.

Market-data calibration and trading use are outside the scope of these computations. The antecedent implementation of Bauer et al. motivates the source-convention and boundary-representation distinctions below; it is not an executable dependency. Scientific attribution is retained independently of software development history.

Figures 7, 9–11, 13 and 14 use the v2.2.0 ensembles and presentation described here. Figures 1–6, 8 and 12 and the two deterministic benchmark tables retain the published v2.1.0 evidence. Figure 8 therefore retains its independently calibrated covariance normalization; it should not be interpreted as a second rendering of the matched estimator in Figure 7b. No physical parameter was refitted to improve visual agreement.

Both books first evolve on one uniform operational grid. Book-specific calendar clocks act only after the complete operational path exists, by selecting the previous completed state. Calendar waits never enter the density recurrence; no calendar interpolation, extrapolation or nonuniform state update is used.

2 Theory-to-computation map

The ordinary build-up kernel is

$$F(x) = 1 - \frac{1 - e^{-x}}{x}, \quad x \geq 0, \quad (1)$$

with $(F(0)=0)$ defined by continuity. The implementation uses `expm1` away from the origin and the local series

$$F(x) = \frac{x}{2} - \frac{x^2}{6} + \frac{x^3}{24} - \frac{x^4}{120} + O(x^5) \quad (2)$$

near zero. Its derivative and local rate elasticity are

$$F'(x) = \frac{1 - (1+x)e^{-x}}{x^2}, \quad S_F(x) = \frac{x F'(x)}{F(x)}. \quad (3)$$

The normalised ordinary components are

$$C_{\text{clock}}(\Delta) = F(\lambda_{12}\Delta), \quad (4)$$

$$C_{\text{coupling}}(\Delta) = F(\kappa\Delta), \quad (5)$$

$$C_{\text{combined}}(\Delta) = C_{\text{clock}}(\Delta)C_{\text{coupling}}(\Delta). \quad (6)$$

The last expression is the paper's leading-order factorisation; the calculation does not promote it to an exact independence result.

For a characteristic time τ , the fractional component is written

$$F_\alpha(\Delta; \tau) = 1 - E_{\alpha,2}[-(\Delta/\tau)^\alpha], \quad 0 < \alpha \leq 1. \quad (7)$$

The release evaluator uses a deterministic real-axis quadrature for $0 < \alpha < 1$, a small-argument series, and the exact ordinary formula at $\alpha = 1$. Its registered release profiles are $\alpha \in \{0.6, 0.8, 1\}$.

The decaying Gaussian source and its selected half-line moment are

$$q_j(y) = -a_j \mu_j y e^{-\mu_j y^2}, \quad M_j^+ = \int_0^\infty y q_j(y) dy = -\frac{a_j \sqrt{\pi}}{4\sqrt{\mu_j}}. \quad (8)$$

The local frozen-front mapping is

$$\kappa_j = \frac{\gamma_{jk} |M_j^+|}{|\mathcal{L}_j|}, \quad \kappa = \kappa_1 + \kappa_2. \quad (9)$$

2.1 Source convention and boundary representation

The translated source uses

$$q_j(y) = -a_j \mu_j y \exp(-\mu_j y^2). \quad (10)$$

The retired antecedent implementation used the distinct expression $\exp[-(\mu_j y)^2]$. These expressions coincide only in special parameter cases and are not interchangeable.

For the numerical density state $\phi_j(x, u)$, the production coupling acts through the receiving front's translation mode

$$T_j(x, u) = -\partial_x \phi_j(x, u). \quad (11)$$

A small multiple of T_j translates the resolved zero crossing to first order. The density profile already supplies the reaction-front width, so this implementation does not add a second fixed selector width or an independent boundary layer. The bounded regularised source in the paper remains an analytical weak/local moment closure. Its correspondence with Eq. (11) is asserted after projection onto front displacement, not as pointwise identity between kernels.

The source convention, fixed operational update, and translation-mode projection are substantive mathematical distinctions from the antecedent formulation. They are tested through component comparisons and a combined fixed-parameter comparison; no pointwise identity between boundary kernels is asserted.

3 Ordinary Epps components and sensitivity

Figure 1 is the analytical baseline used to interpret the simulation results. The three panels nondimensionalise the clock rate to one and vary the response-to-clock rate ratio. The curves separate the two mechanisms before forming the combined product.

The short-scale limit of each component is linear in its dimensionless argument, so its rate elasticity approaches one. Once the component reaches its plateau, the elasticity approaches zero. For the product, the partial log-sensitivity to each rate is inherited from that component, while a common proportional perturbation of both rates has the sum of the two partial elasticities.

The combined ordinary curve is invariant under exchange of the clock and response rates. Moreover,

$$C_{\text{combined}}(\Delta) = \frac{\lambda_{12}\kappa}{4}\Delta^2 + O(\Delta^3) \quad (12)$$

at short scale. Thus the leading combined curve alone reveals the rate product rather than assigning each timescale to its mechanism. The simulation results below therefore retain independently declared refresh and response rates.

4 Fractional order sensitivity

From Eq. (7),

$$F_\alpha(\Delta; \tau) = \frac{(\Delta/\tau)^\alpha}{\Gamma(\alpha + 2)} + O(\Delta^{2\alpha}). \quad (13)$$

A combined fractional curve therefore has leading power $\Delta^{\alpha_c + \alpha_r}$. Equal-sum order pairs are deliberately included in Figure 3 to show the resulting local confounding.

5 Conditional reaction-boundary propagation

Equation (9) yields conditional log-elasticities $+1, +1, -1/2, -1$ with respect to coupling strength, source amplitude, source width, and front-slope magnitude. Figure 4 changes one quantity at a time. This is a propagation experiment: multiple structural parameter combinations generate the same effective response rate and hence the same analytic Epps curve.

Three distinct notions must not be collapsed into a single “boundary thickness”: the local reaction-front slope $|\mathcal{L}_j|$, the continuum selector regularisation ε , and the lattice spacing Δx . The first enters the structural response mapping. The second regularises side selection. The third is a numerical resolution parameter.

6 Continuum selector and finite-grid representation

The paper’s smooth selector is

$$W(y, z; \varepsilon) = \frac{1}{2} \left[1 + \tanh \left(\frac{yz}{\varepsilon} \right) \right]. \quad (14)$$

For a centred symmetric full domain, $yq_j(y)$ is even and the hyperbolic-tangent contribution is odd. Consequently the selected first moment is independent of ε in the continuum calculation. Finite domains, misaligned numerical selectors, and coarse grids can break this cancellation. Figure 5 diagnoses those representation effects and propagates them through an effective response rate.

Ordinary Epps components under separated clock and coupling timescales

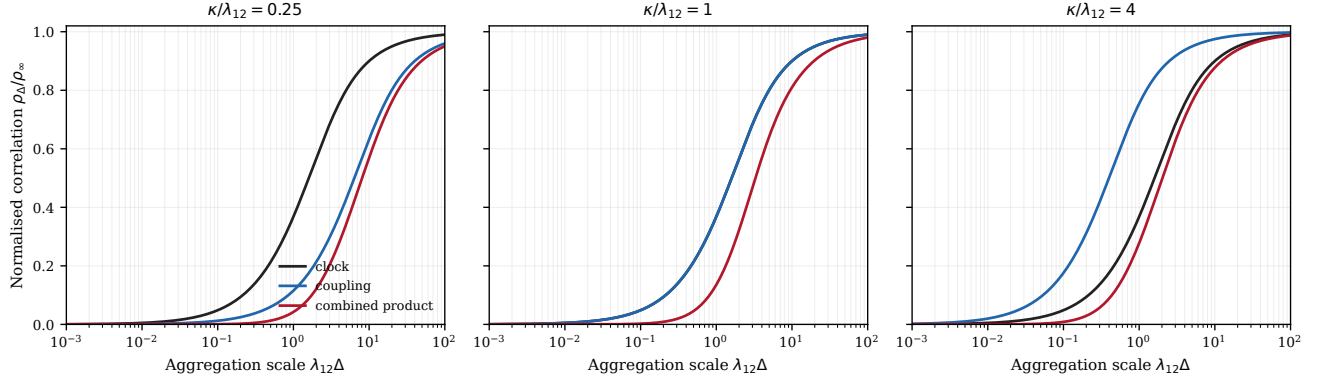


Figure 1: **Ordinary Epps components under separated clock and coupling timescales.** The clock curve is $F(\lambda_{12}\Delta)$, the coupling-response curve is $F(\kappa\Delta)$, and the combined curve is their separable product. The horizontal axis is aggregation scale in clock characteristic-time units and the vertical axis is ρ_Δ/ρ_∞ . Panels use $\kappa/\lambda_{12} = 0.25, 1, 4$, with $\lambda_{12} = 1$ and $\rho_\infty = 1$. The figure shows how the mechanisms contribute across separated and comparable timescales and supplies the analytic component baseline. It supports the component decomposition under the paper’s renewal-envelope and separability assumptions; it does not fit market data, simulate an order book, prove statistical independence, or establish the approximation outside those assumptions.

Aggregation-scale sensitivity of ordinary Epps components

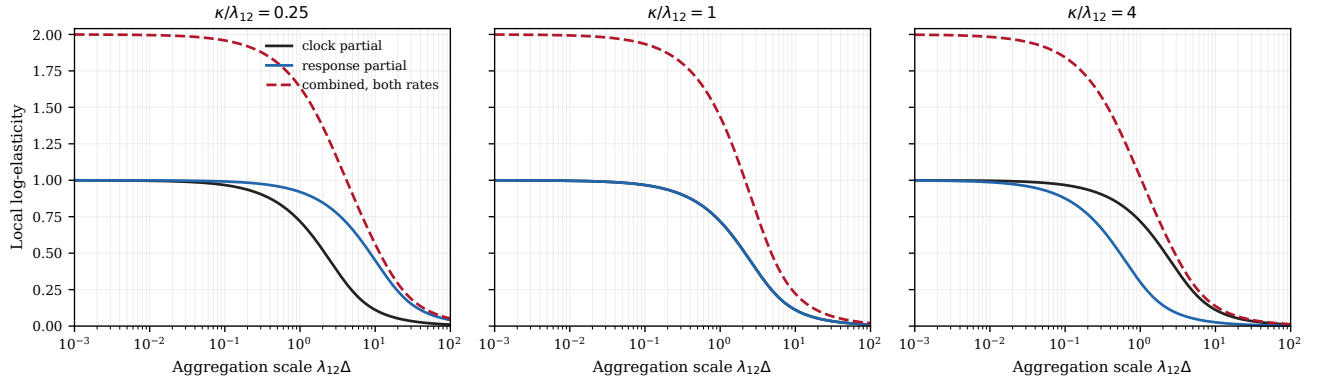


Figure 2: **Aggregation-scale sensitivity of ordinary Epps components.** Curves report $S_F(x) = xF'(x)/F(x)$ for clock refresh and coupling response. The dashed red curve is the log-sensitivity of the combined product when both rates receive the same proportional perturbation. Panels use the Figure 1 rate ratios. Sensitivity tends to one per component in the resolved short-scale limit and to zero on the long-scale plateau, locating the aggregation range that contains rate information. The figure supports local, model-conditional sensitivity statements; it is not a global-identifiability proof, an empirical uncertainty estimate, or evidence that either rate can be recovered without adequate short-scale observations.

Fractional Epps curves and short-scale exponent sensitivity

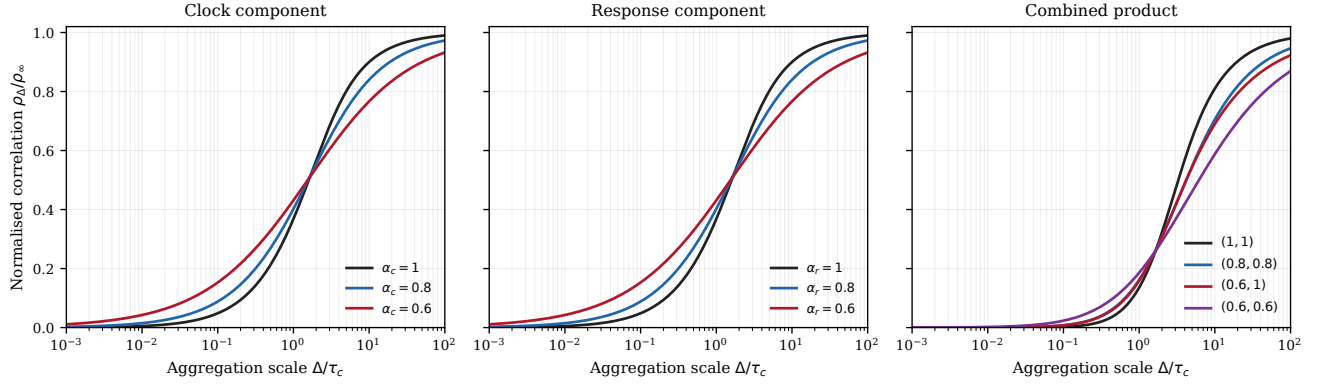


Figure 3: **Fractional Epps curves and short-scale exponent sensitivity.** Clock and response components use Eq. (7), with unit characteristic times and the diagnosed evaluator recorded in the numerical-design note; $\alpha = 1$ recovers the ordinary kernel. Combined curves use order pairs $(1, 1)$, $(0.8, 0.8)$, $(0.6, 1)$, $(0.6, 0.6)$. The equal-sum cases $(0.8, 0.8)$ and $(0.6, 1)$ share leading short-scale power $\Delta^{1.6}$, exposing local confounding even when later-scale curvature differs. The figure shows formula sensitivity to fractional order and the ordinary limit. It does not identify both orders from a combined curve alone, attach an unconditional “slower” interpretation across dimensional conventions, or validate a fractional market-clock mechanism empirically.

Conditional propagation of reaction-boundary structure into the Epps curve

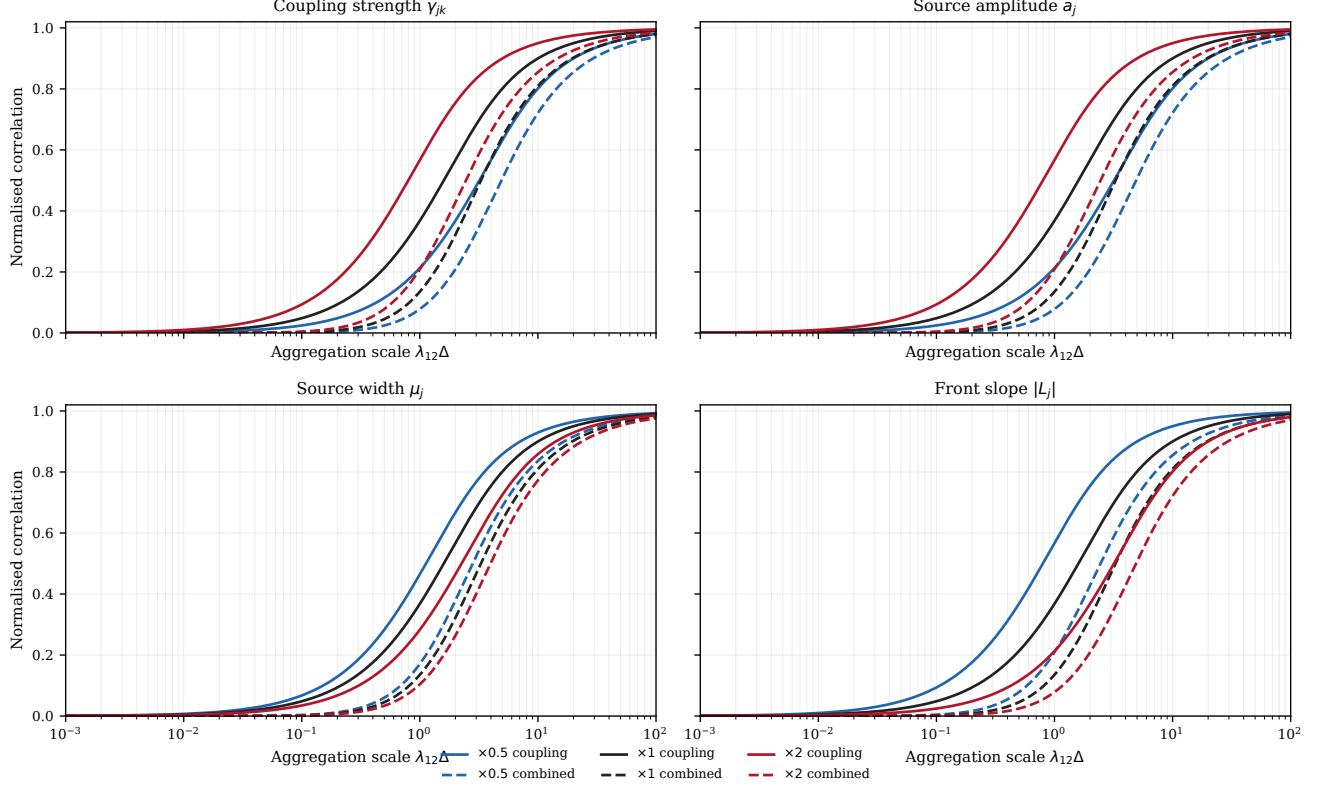


Figure 4: **Conditional propagation of reaction-boundary structure into the Epps curve.** The response rate is varied through Eqs. (8)–(9). Two symmetric books, unit source amplitude, width and front-slope magnitude, and $\gamma_{jk} = 2/\sqrt{\pi}$ give baseline total response rate $\kappa = 1$. Each panel changes one input by factors 0.5, 1, 2, holding the others and the unit clock rate fixed; solid curves are coupling factors and dashed curves their combined products with the clock factor. This supports conditional propagation through the analytic response mapping. It does not uniquely infer reaction-front thickness, source width, or coupling strength from an Epps curve and does not claim pointwise equivalence to the legacy lattice coupling.

Finite-grid boundary representation and induced Epps-curve distortion

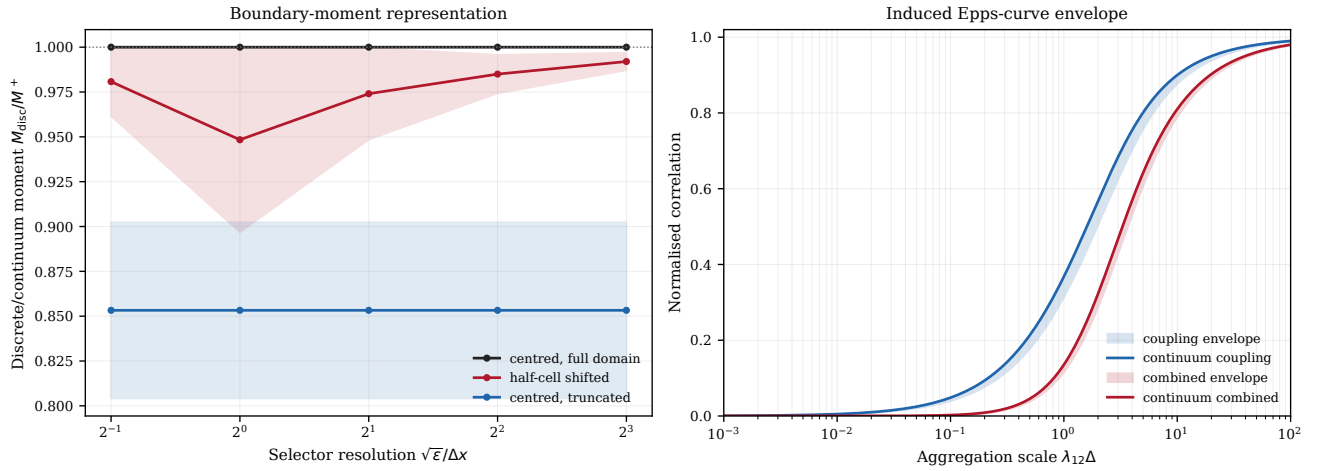


Figure 5: **Finite-grid boundary representation and induced Epps-curve distortion.** The left panel reports discrete-to-continuum first-moment ratios as selector resolution $\sqrt{\epsilon}/\Delta x$, lattice spacing, selector alignment, and domain margin vary. Lines give the midpoint and bands the range over four lattice spacings. The centred full-domain case is the parity control; half-cell selector displacement and truncation measure numerical representation errors. The right panel propagates all diagnosed effective-rate ratios through the coupling and combined Epps formulas; lines are continuum baselines and bands are discrete-representation envelopes. Structural parameters are fixed. The figure supports convergence and numerical-sensitivity statements and defines resolution metadata required for v2 overlays. It does not introduce a new continuum Epps mechanism or treat selector width, lattice spacing, and reaction-front thickness as interchangeable.

7 Calendar-time representation and memory diagnostic

The aggregation variable in Figures 1–5 remains the observation-window lag, but is reported relative to the clock characteristic time. Writing

$$x = \lambda_{12}\Delta t = \frac{\Delta t}{\tau_c}, \quad \tau_c = \lambda_{12}^{-1}, \quad (15)$$

shows that a calendar-time presentation requires only a declared time convention. For an ordinary exponential memory with rate r ,

$$h_r(u) = e^{-ru}, \quad F(r\Delta t) = 1 - \frac{1}{\Delta t} \int_0^{\Delta t} h_r(u) du. \quad (16)$$

For the clock mechanism, $h_{\lambda_{12}}$ is the no-refresh survival. For the coupling mechanism, h_{κ} is the relaxation survival and equals the normalised positive-lag exponential covariance kernel used in the paper. Figure 6 therefore presents the same analytical factors as Figure 1, not a new Epps mechanism.

The distinction from the simulation output is substantive. A sample autocorrelation requires a generated path, a declared estimator and finite-sample uncertainty. The deterministic inset exposes only the memory functions that enter Eq. (16). Figure 11 reports the separate path-derived log-mid increment and trade-sign autocorrelations without retrospectively relabelling this analytical diagnostic.

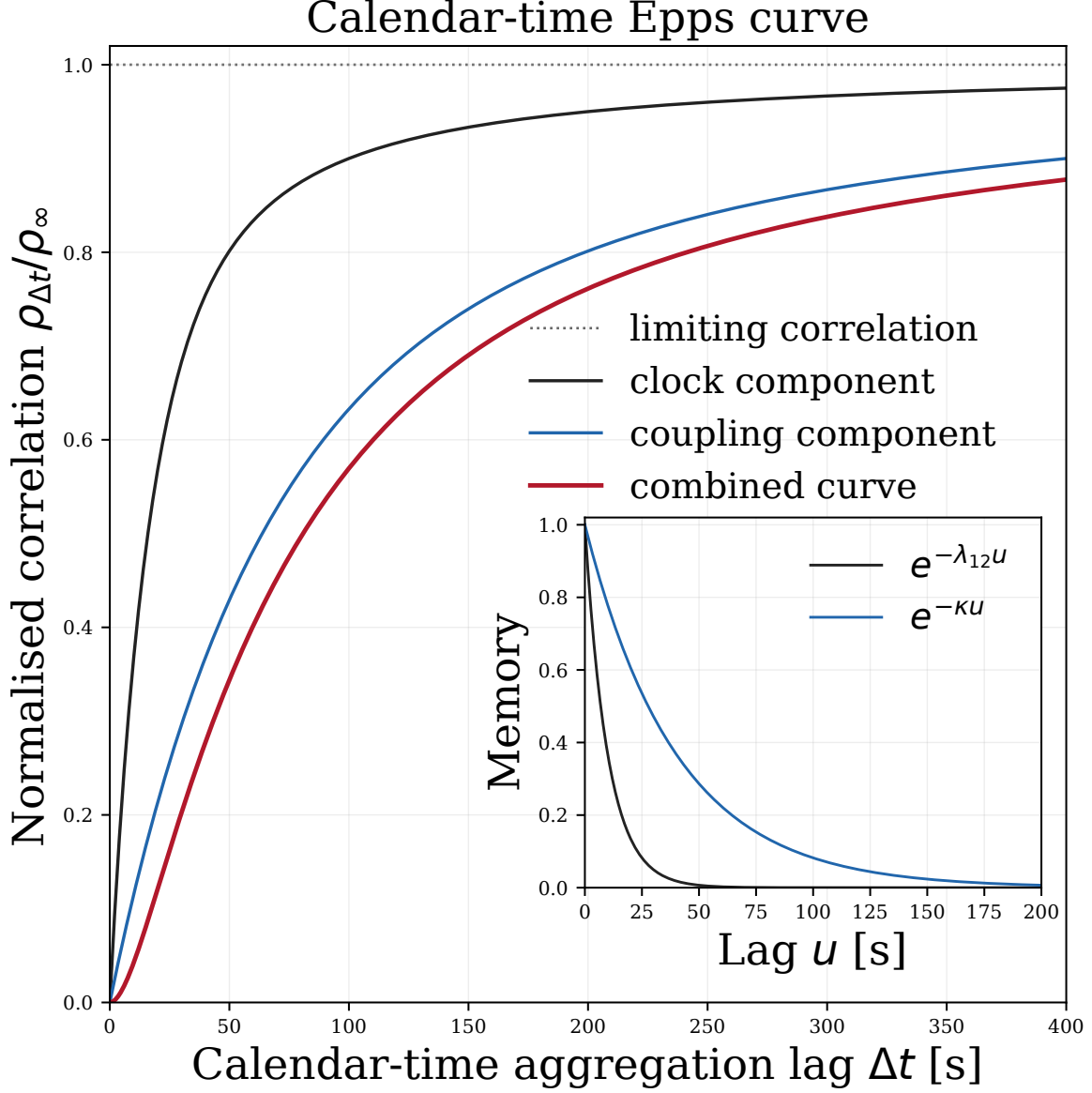


Figure 6: **Calendar-time Epps curve.** The square main panel rewrites the Figure 1 slow-response profile on a linear calendar-time axis using the explicitly illustrative convention $\lambda_{12} = 0.1 \text{ s}^{-1}$, $\kappa = 0.025 \text{ s}^{-1}$, and $\rho_{\infty} = 1$, corresponding to $\tau_c = 10 \text{ s}$ and $\tau_r = 40 \text{ s}$. Black, blue and red curves are the clock factor, coupling factor and their leading-order separable product; the grey dotted line marks the normalised limiting correlation $\rho_{\Delta t}/\rho_{\infty} = 1$. The enlarged inset reports $h_c(u) = e^{-\lambda_{12}u}$ and $h_r(u) = e^{-\kappa u}$: respectively the no-refresh survival and coupling-relaxation survival, with the latter also equal to the normalised positive-lag exponential covariance kernel. The seconds scale is not fitted to data, and the inset is neither a path autocorrelation nor a power-spectrum estimate. Path-derived autocorrelations are reported separately in Figure 11.

8 Uniform-operational simulation

Let $u_n = n\Delta u$ be the common operational grid. Both density fields are advanced on this grid before any calendar observation is constructed. External innovations are book-specific, the translation-mode coupling uses Eq. (11), and labelled order events modify the density state at declared operational steps. The boundary price is the accepted unique interior zero crossing of the signed density field.

For book j , let $\tau_{j,m}$ be its completed calendar refresh events. The software first selects the latest event at or before t , then the latest uniform operational node at or before that event:

$$m_j(t) = \max\{m : \tau_{j,m} \leq t\}, \quad n_j(t) = \max\{n : u_n \leq \tau_{j,m_j(t)}\}, \quad P_j^{\text{cal}}(t) = P_j^{\text{op}}(u_{n_j(t)}). \quad (17)$$

Equation (17) is previous-refresh subordination. It selects an already completed state and never changes an operational update interval.

8.1 Executable numerical algorithms

Algorithms 1–4 make the software’s two time layers explicit. Operational evolution uses one frozen pre-step state; renewal generation and calendar observation act only on completed paths. Every coupled update reads the same frozen histories, and every shocked/control comparison uses the same realised clock.

Algorithm 1 Coupled operational-time reaction-boundary evolution

- Input:** Uniform grid x_i , step Δu , histories $\{\varphi_m^{i(j)}\}_{m \leq n}$, and boundaries $p_n^{(j)}$
Input: Sources $s_n^{i(j)}$, optional additive fields $\delta_n^{i(j)}$, and directed rates $\kappa_{jk} \geq 0$
Output: Next densities and boundaries, coupling/source fields, and diagnostics
- 1: Apply any declared market-order consumption map once to $\varphi_n^{(j)}$; retain the pre-event boundaries
 - 2: Freeze the post-consumption $\varphi_{n,\text{pre}}^{i(j)} \leftarrow \varphi_n^{i(j)}$ and $p_{n,\text{pre}}^{(j)} \leftarrow p_n^{(j)}$ for every book
 - 3: **for all** ordered pairs $j \leftarrow k$ **do**
 - 4: $z_{jk,n} \leftarrow p_{n,\text{pre}}^{(j)} - p_{n,\text{pre}}^{(k)}$
 - 5: $v_{j,n}^{\Delta x} \leftarrow -\partial_x^{\Delta x} \varphi_{n,\text{pre}}^{(j)}$; set its outer values to zero
 - 6: $\ell_{T,n}^{i(j,k)} \leftarrow -\kappa_{jk} z_{jk,n} v_{j,n}^{\Delta x}(x_i)$
 - 7: **for all** books j **do**
 - 8: $\ell_{T,n}^{i(j)} \leftarrow \sum_{k \neq j} \ell_{T,n}^{i(j,k)}$
 - 9: $c_n^{i(j)} \leftarrow s_n^{i(j)} + \delta_n^{i(j)} + \ell_{T,n}^{i(j)}$
 - 10: $\varphi_{n+1,\text{raw}}^{(j)} \leftarrow \mathcal{U}_{\Delta u}^{(j)}[\{\varphi_m^{(j)}\}_{m \leq n}, c_n^{(j)}]$
 - 11: Impose $\varphi_{n+1}^{0(j)} = \varphi_{n+1}^{I(j)} = 0$
 - 12: Form the admissible exact-zero and strict-crossing set $\mathcal{Z}_{n+1}^{(j)}$ defined below
 - 13: Require $\mathcal{Z}_{n+1}^{(j)} \neq \emptyset$ and set $p_{n+1}^{(j)} \leftarrow \mathcal{B}_j[\mathcal{Z}_{n+1}^{(j)}; p_n^{(j)}]$
 - 14: **return** states, boundaries, candidate data, fields, sources, and recurrence contributions
-

For market-order runs, density is consumed before the completed update, with the pre-event boundary retained for the source displacement. The coupling gradient is evaluated on the post-consumption density; the removed quantity is not injected again as an additive source. The optional additive-field term excludes

any market-order consumption already applied before the update. These ordering details determine the implementation used in Figures 9–14.

Here $\mathcal{U}_{\Delta u}^{(j)}$ is the fixed-step DTRW recurrence; the additive field $c_n^{(j)}$ enters its source channel rather than forming a separate update. The boundary candidates used in the algorithm are

$$\mathcal{Z}_{n+1,\text{exact}}^{(j)} = \left\{ x_i : 1 \leq i \leq I-1, \varphi_{n+1}^{i(j)} = 0, \varphi_{n+1}^{i-1(j)} \varphi_{n+1}^{i+1(j)} < 0, |L_i^{(j)}| \geq L_{\min} \right\}, \quad (18)$$

$$\mathcal{Z}_{n+1,\text{cross}}^{(j)} = \left\{ x_i - \frac{\varphi_{n+1}^{i(j)}}{L_{i+1/2}^{(j)}} : \varphi_{n+1}^{i(j)} \varphi_{n+1}^{i+1(j)} < 0, |L_{i+1/2}^{(j)}| \geq L_{\min} \right\}, \quad (19)$$

with

$$L_i^{(j)} = \frac{\varphi_{n+1}^{i+1(j)} - \varphi_{n+1}^{i-1(j)}}{2\Delta x}, \quad L_{i+1/2}^{(j)} = \frac{\varphi_{n+1}^{i+1(j)} - \varphi_{n+1}^{i(j)}}{\Delta x}. \quad (20)$$

The full set is their union. Unique selection requires one candidate; nearest-previous selection requires a unique minimum distance from $p_n^{(j)}$ and rejects distance ties. Either $\kappa_{jk} = 0$ or $z_{jk,n} = 0$ makes the corresponding directed field exactly zero.

Algorithm 2 Book-specific previous-refresh calendar observation

- Input:** Completed prices $p^{(j)}(u_n)$ on a uniform operational grid
Input: Strict refresh paths $0 = \tau_{j,0} < \dots < \tau_{j,M_j}$, nondecreasing queries t_r , and integer lags h_q
Output: Calendar prices, index/support records, pooled components, and optional conditional moments
- 1: Require $0 \leq t_0 \leq \dots \leq t_R \leq \min\{u_N, \tau_{1,M_1}, \tau_{2,M_2}\}$
 - 2: **for all** books j and queries t_r **do**
 - 3: $m_j(t_r) \leftarrow \max\{m : \tau_{j,m} \leq t_r\}$
 - 4: $n_j(t_r) \leftarrow \max\{n : u_n \leq \tau_{j,m_j(t_r)}\}$
 - 5: $P_{\Delta}^{(j)}(t_r) \leftarrow p^{(j)}(u_{n_j(t_r)})$
 - 6: **for all** lags h_q and admissible origins t_r **do**
 - 7: Form $R_{q,r}^{(j)} \leftarrow P_{\Delta}^{(j)}(t_{r+h_q}) - P_{\Delta}^{(j)}(t_r)$
 - 8: Accumulate $R_{q,r}^{(1)} R_{q,r}^{(2)}$ and $(R_{q,r}^{(j)})^2$ into S_q^{12} and S_q^{jj}
 - 9: Record $J_{j,q,r} = (a_j, b_j]$ and overlap $\Theta_{q,r}$
 - 10: **if** the symmetric reduced conditional reference is requested **then**
 - 11: Evaluate $K_{q,r}$, $C_{q,r}^{12}$, and $V_{q,r}^{(j)}$ below
 - 12: Pool component sums across path groups, then compute $\hat{\rho}_q = S_q^{12} / \sqrt{S_q^{11} S_q^{22}}$
 - 13: Form delete-one-group ratios and the path-group jackknife error
 - 14: **return** observations, indices, support, components, ratios, and uncertainty
-

The nested map in lines 2–4 is the exact previous-completed-state inverse: it first selects a refresh event and then an operational node, without interpolation, extrapolation, or terminal-state clamping. For each return,

$$(a_1, b_1, a_2, b_2) = \Delta u (n_1(t_r), n_1(t_{r+h_q}), n_2(t_r), n_2(t_{r+h_q})), \quad \Theta_{q,r} = \max\{0, \min(b_1, b_2) - \max(a_1, a_2)\}. \quad (21)$$

When requested, the symmetric conditional reference uses $\kappa = \kappa_{12} + \kappa_{21} > 0$ and

$$K_{q,r} = e^{-\kappa|b_1-b_2|} - e^{-\kappa|b_1-a_2|} - e^{-\kappa|a_1-b_2|} + e^{-\kappa|a_1-a_2|}, \quad (22)$$

$$C_{q,r}^{12} = \Theta_{q,r} - \frac{K_{q,r}}{2\kappa}, \quad (23)$$

$$V_{q,r}^{(j)} = (b_j - a_j) - \frac{e^{-\kappa(b_j-a_j)} - 1}{\kappa}. \quad (24)$$

The final minus sign is intentional: $e^{-\kappa d} - 1 \leq 0$, so the Ornstein–Uhlenbeck spread contribution adds to marginal variance and subtracts from cross-covariance. Identity clocks recover the implemented coupling-only moments to the registered tolerance.

For G independent path groups, uncertainty uses

$$\hat{\rho}_q^{(-g)} = \frac{S_q^{12} - S_{gq}^{12}}{\sqrt{(S_q^{11} - S_{gq}^{11})(S_q^{22} - S_{gq}^{22})}}, \quad \text{se}_{\text{JK}}(\hat{\rho}_q) = \left[\frac{G-1}{G} \sum_{g=1}^G \left(\hat{\rho}_q^{(-g)} - \bar{\rho}_q^{(-)} \right)^2 \right]^{1/2}. \quad (25)$$

Algorithm 3 Mittag–Leffler and exponentially tempered renewal paths

Input: Order $0 < \beta \leq 1$, scale $\tau > 0$, horizon $T > 0$, and independent open uniforms (U_r, V_r, E_r, A_r) supplied by the caller

Input: Optional tempering rate $\lambda_T > 0$

Output: Strict renewal events $0 = \tau_0 < \dots < \tau_M$ with $\tau_M \geq T$

- 1: **for** $r = 1, 2, \dots$ until the retained cumulative wait reaches T **do**
- 2: **if** $\beta = 1$ **then**
- 3: $S_r \leftarrow 1$
- 4: **else**
- 5: $\theta_r \leftarrow \pi U_r$ and $Z_r \leftarrow -\log V_r$
- 6: $S_r \leftarrow \frac{\sin(\beta\theta_r)}{\sin(\theta_r)^{1/\beta}} \left[\frac{\sin((1-\beta)\theta_r)}{Z_r} \right]^{(1-\beta)/\beta}$
- 7: $W_r \leftarrow \tau(-\log E_r)^{1/\beta} S_r$
- 8: **if** untempered or $A_r \leq e^{-\lambda_T W_r}$ **then**
- 9: Append W_r and its cumulative sum to the renewal path
- 10: **return** event times, retained waits, law parameters, stream id, and supported horizon

Algorithm 3 is exact for the declared Laplace transforms in Eqs. (37) and (38). It neither clips heavy waits nor rescales the accepted sample to a target mean. At $\beta = 1$ the untempered construction reduces to exponential waiting intervals with mean τ .

Algorithm 4 Common-input paired impact under an observation clock

Input: Completed control paths $P_{p,j}^0(u_n)$ and signed-event shocked paths $P_{p,b,s,j}^1(u_n)$ generated from identical initial states and inputs

Input: One book-specific renewal pair per path, event step n_e , and lags ℓ_q on the query grid

Output: Own/cross impact means, independent-group standard errors, and active fractions

- 1: **for all** paths p **do**
 - 2: Apply Algorithm 2 to the control path and retain its selected indices $n_{p,j}(t)$
 - 3: **for all** event books b and signs $s \in \{-1, +1\}$ **do**
 - 4: Apply the *same* realised clock pair to the shocked path
 - 5: **for all** response books j and lags ℓ_q **do**
 - 6: $I_{p,b,s,j}(\ell_q) \leftarrow s[P_{p,b,s,j}^{1,c}(t_e + \ell_q) - P_{p,j}^{0,c}(t_e + \ell_q)]$
 - 7: $A_{p,j}(\ell_q) \leftarrow \mathbf{1}\{n_{p,j}(t_e + \ell_q) \geq n_e\}$
 - 8: Average signs, symmetry partners and clock replicates within each independent primitive group; retain directional book pairs where required and compute errors across groups
 - 9: **return** impact curves and active-observation fractions
-

For a meta-order, line 7 is evaluated at each child step and at declared lags after the final child; activity at relaxation lag is measured relative to the final child. An inactive observation contributes its actual paired value, normally zero before the event is observed, rather than being removed by conditioning.

Algorithm 1 owns dynamics and boundary extraction. Algorithms 2 and 3 own observation only, while Algorithm 4 owns the estimand. This separation is the principal control for the Poisson, Mittag–Leffler and tempered observation-clock comparisons.

8.2 Theory-to-simulation Epps comparison

Figure 7 in this supplement corresponds to Figure 1 in the paper. Its combined three-square layout retains common axis limits and component colours; separate square exports are also supplied. Each comparison uses 512 independent validation paths, 8000 seconds after 200 seconds of dynamical warm-up, $\Delta x = 0.1$ and a 0.5-second update. The clock-only and combined panels use four book-specific Poisson-clock replicates per path. A replicate is not an additional independent operational path. The existing 256-path independent calibration ensembles remain available as diagnostics.

The normalization is explicit. In Figure 7a the plotted attenuation is the ratio of asynchronous to synchronous pooled return correlations on the *same* validation ensemble:

$$\hat{A}(\Delta) = \frac{\hat{\rho}_{\text{async}}(\Delta)}{\hat{\rho}_{\text{sync}}(\Delta)}. \quad (26)$$

In Figures 7b and 7c let $X = (P_1 + P_2)/2$. The plotted normalized covariance is

$$\hat{B}_c(\Delta) = \frac{\sum_{g,r} \overline{R_{1,g,r}^{(c)} R_{2,g,r}^{(c)}}}{\sum_{g,r} (R_{X,g,r}^{\text{op}})^2}, \quad (27)$$

where g denotes an independent path, r an admissible origin, and the overbar averages clock replicates where applicable. It is a covariance response, not Pearson correlation. These matched estimators differ from dividing by an independently estimated long-scale covariance rate times Δ in the published comparison. Matching retains numerator–denominator dependence and reduces normalization noise; it is not a fit of a response rate or a time map.

Shading in all three panels is the approximate pointwise 98% interval $\hat{\theta} \pm t_{0.99,511} \text{se}_{\text{JK}}(\hat{\theta})$, using a delete-one-independent-path jackknife that recomputes the complete ratio. The critical multiplier is approximately 2.334. These intervals describe uncertainty in the estimated mean/pooled statistic, not the spread of individual paths, simultaneous coverage over all lags, or discretization/model error. The increase from 95% to 98% widens them by about 19%; there is no minimum graphical width. A narrow band may remain largely hidden by coincident curves at the printed scale. Black reference strokes are wider than the coloured simulation strokes so that both remain noticeable at overlap.

The black clock curve is the equal-rate previous-refresh factor. The black combined curve remains the paper's product approximation. The dashed grey curve is the stationary joint reduced reference in Eq. (30). The reduced conditional moments at the realized refresh indices are retained in the supplied numerical comparison; neither reference is an exact identity for the full reaction–diffusion dynamics. No smoothing of simulated means, selected-seed replacement, or parameter refitting is applied.

Why the combined curve differs from the product. Clock-only attenuation and coupling-induced covariance describe different operations. The clock selects observation times; coupling spreads cross-book dependence over time. Applying independent clocks to a coupled process therefore requires averaging that time-dependent covariance over the sampled intervals. It is not equivalent to multiplying the two separately averaged component curves.

This distinction can be calculated explicitly for the stationary reduced model underlying the grey reference. Write $P_1 = X + Z/2$ and $P_2 = X - Z/2$, with independent Brownian pair centre X and stationary Ornstein–Uhlenbeck spread Z . Let C be the variance rate of X , and use the spread variance $2C/\kappa$ of the symmetric closure. The cross-covariance kernel of price increments is

$$K_{12}(s) = \frac{C\kappa}{2} e^{-\kappa|s|}. \quad (28)$$

For independent stationary Poisson observation clocks, each of rate $\lambda = \lambda^{\text{clk}}$ and independent of the prices, previous-refresh observation convolves this cross-kernel with the difference-of-clock-age density $h_\lambda(s) = (\lambda/2)e^{-\lambda|s|}$. This is a cross-book result; it does not apply the same convolution to a book's marginal variance. Integrating the resulting kernel over a return window gives

$$G(\Delta) = \frac{2}{C\Delta} \int_0^\Delta (\Delta - s)(K_{12} * h_\lambda)(s) ds \quad (29)$$

$$= \frac{\lambda^2 F(\kappa\Delta) - \kappa^2 F(\lambda\Delta)}{\lambda^2 - \kappa^2}, \quad \lambda \neq \kappa. \quad (30)$$

At equal rates the continuous limit is $G(\Delta) = F(x) - xF'(x)/2$, with $x = \kappa\Delta$. Equation (30) is a population covariance normalized by $C\Delta$. It is the stationary continuous-time counterpart of the reduced conditional calculation, not an exact solution of the full reaction–diffusion dynamics or a finite-sample ratio identity.

The short-time powers expose the difference:

$$G(\Delta) = \frac{\lambda\kappa}{2(\lambda + \kappa)} \Delta + O(\Delta^3), \quad F(\lambda\Delta)F(\kappa\Delta) = \frac{\lambda\kappa}{4} \Delta^2 + O(\Delta^3). \quad (31)$$

The joint reduced response starts linearly, whereas the product starts quadratically. Thus the paper's product approximation is not the short-time asymptotic of this joint reduced model. With the fixed rates $\lambda = 0.1$ and $\kappa = 0.025 \text{ s}^{-1}$, at 20 seconds the joint expression is 0.18942, the product is 0.12095, and the pooled simulation is 0.18717. Increasing the ensemble can expose this approximation residual more clearly, but cannot remove it. Both analytical expressions tend to one as the aggregation lag increases.

The mean clock wait is 10 seconds and the coupling relaxation time is 40 seconds. The earlier Figure 7 curves began at 20 seconds, where the clock factor has already reached 0.568. A log-log change of axes alone cannot recover the missing initial rise. The short-lag extension therefore evaluates all three panels below 20 seconds and checks the numerical grid separately from sampling precision. The algebra above was checked against direct kernel convolution and return-window quadrature without parameter fitting.

Short-lag numerical check. The same 512-path ensembles were replayed at eight additional lags: 0.5, 1, 2, 3, 5, 7.5, 10 and 15 seconds. Their previously saved moments at 20–400 seconds were recovered before the additional lags were used. At 0.5 seconds, the clock-only estimate is 0.02460 ± 0.00064 , compared with $F(0.05) = 0.02459$, and the combined estimate is 0.00531 ± 0.00075 , compared with $G(0.5) = 0.00500$; the quoted uncertainties are pointwise 98% half-widths.

The coupling-only value at that lag is -0.00638 ± 0.00167 , compared with the continuous reduced value 0.00622. Here the return contains only one 0.5-second update. The explicit discrete reduced closure explains why this distinction matters. With update h , spread coefficient $q = 1 - \kappa h$ and $\Delta = mh$, its stationary normalized synchronous covariance is

$$B_h(\Delta) = 1 - \frac{1 - (1 - \kappa h)^m}{\kappa \Delta (1 - \kappa h/2)}, \quad 0 < \kappa h < 2. \quad (32)$$

At one update, $B_h(h) = -\kappa h/(2 - \kappa h)$, which is -0.00629 here. The two coupling drifts oppose one another, while a new independent shock reaches the other book through subsequent updates. This discrete result is a reduced benchmark, not an exact identity for the full density solver. As $h \rightarrow 0$ at fixed Δ , it tends to $F(\kappa \Delta)$.

A paired 64-path, 2000-second screen compared $(\Delta x, h) = (0.1, 0.5 \text{ s})$ and $(0.05, 0.125 \text{ s})$, preserving the diffusion scaling and using nested innovations and the same continuous observation clocks. At the 0.5-second lag the coupling-only change is $+0.00938 \pm 0.00009$; the clock-only and combined changes are -0.00047 ± 0.00091 and $+0.00001 \pm 0.00113$. Thus sampling precision, discrete updating and product approximation have different roles. The main figure retains the declared 512-path production grid, and the paired refinement results are summarized above. The smallest-lag coupling-only points are not continuum-converged; the required scale ordering is $h \ll \Delta$. Nonpositive estimates and interval endpoints remain visible on the linear axes and are omitted from the log-log axes.

Comparison with the published Figure 7. The v2.1.0 shading represents pointwise 95% jackknife uncertainty in the estimated simulation statistic, not the distribution of individual sample paths. In panel (c), the covariance was divided by a frozen, independently calibrated covariance rate times the aggregation lag. The plotted standard errors condition on that frozen rate. The v2.2.0 comparison uses the matched pair-centre denominator in Eq. (27), 512 validation paths and longer records, retaining numerator–denominator dependence. Consequently the change in envelope width cannot be attributed to path count alone.

Across the 20 common lags from 20 to 400 seconds, the old combined 95% interval has maximum half-width 0.19958; the new matched 98% interval has maximum half-width 0.00717. Recomputing the new ensemble with independent calibration held fixed gives maximum 98% half-width 0.02807, while changing the mean by at most 0.00151 relative to the matched estimator. This isolates a substantial normalization contribution to precision on the same new ensemble; it does not measure uncertainty in the calibration rate itself.

The black leading-order product is inside the published 95% intervals at 18 of 20 lags, whereas the grey same-clock conditional reference is inside at all 20. At $\Delta = 20 \text{ s}$, the published simulation is 0.20416, the product is 0.12095, and the interval half-width is 0.03423: even the published band excludes the product

there. In the new comparison the product is outside the 98% intervals at all 20 lags, while the conditional reference remains inside all 20. These are descriptive counts across correlated lags, not simultaneous coverage tests. The increased precision exposes the distinction between a product of component factors and the conditional calculation using the actual refresh indices. It does not establish an exact identity between the reduced reference and full reaction–diffusion dynamics. Enlarging the ensemble cannot remove a systematic approximation residual.

8.3 Translation-mode component

The translation-mode coupling is tested separately under identity clocks before it is used in the combined route. Deterministic signed-front relaxations diagnose the registered response rate; stochastic holdout paths test normalized covariance and the distinct finite-scale return correlation. The covariance scale is frozen from a disjoint calibration ensemble.

9 Event impact and dependence diagnostics

Market-order events consume opposing density at the active boundary and are recorded on an immutable event tape. Impact is measured by pairing each event path with an otherwise identical zero-input control. Responses are aggressor-signed log-boundary differences, so own and cross effects use one common linear scale. Calendar versions are obtained only after the operational paths are complete, using Eq. (17).

A meta-order is a declared schedule of child market orders with fixed total quantity and book. Equal-volume horizon comparisons distinguish schedule timing from a true participation rate, which would require background market-order volume not present in this model.

For a sequence X_n , let $\bar{X}_{L,k}$ and $\bar{X}_{R,k}$ denote the separate sample means of $\{X_n\}_{n=1}^{N-k}$ and $\{X_{n+k}\}_{n=1}^{N-k}$. The registered positive-lag statistic is the Pearson correlation of those two overlapping slices,

$$\hat{\rho}_X(k) = \frac{\sum_{n=1}^{N-k} (X_n - \bar{X}_{L,k})(X_{n+k} - \bar{X}_{R,k})}{\sqrt{\left[\sum_{n=1}^{N-k} (X_n - \bar{X}_{L,k})^2\right] \left[\sum_{n=1}^{N-k} (X_{n+k} - \bar{X}_{R,k})^2\right]}}, \quad \hat{\rho}_X(0) = 1. \quad (33)$$

The price statistic is applied to log-mid increments, never to price levels. Trade-sign diagnostics distinguish ground-truth aggressor signs, a quote/midpoint rule and the historically inherited tick rule. Event-time signs and calendar-binned signed flow are separate estimands.

9.1 Fixed-time order-book shock recovery

Figure 12 follows one buy market order in book 1 through the current fixed-grid operational dynamics. Immediately before the declared operational step, the order consumes opposing density from the reaction boundary outwards. If M_{1,n_*} denotes the resulting density increment, then

$$\varphi_{1,n_*}^+ = \varphi_{1,n_*}^- + M_{1,n_*}, \quad \Delta x \sum_i |M_{1,n_*,i}| = Q. \quad (34)$$

The event is applied once and only to book 1. The second book remains active through the receiving-front translation-mode coupling, although its density and the coupling contribution are not separately drawn.

For a completed operational step the stored density satisfies

$$\varphi_{j,n+1} = H_{j,n} + \varphi_{j,n}^+ + C_{j,n} + A_{j,n} + L_{j,n} + B_{j,n}, \quad (35)$$

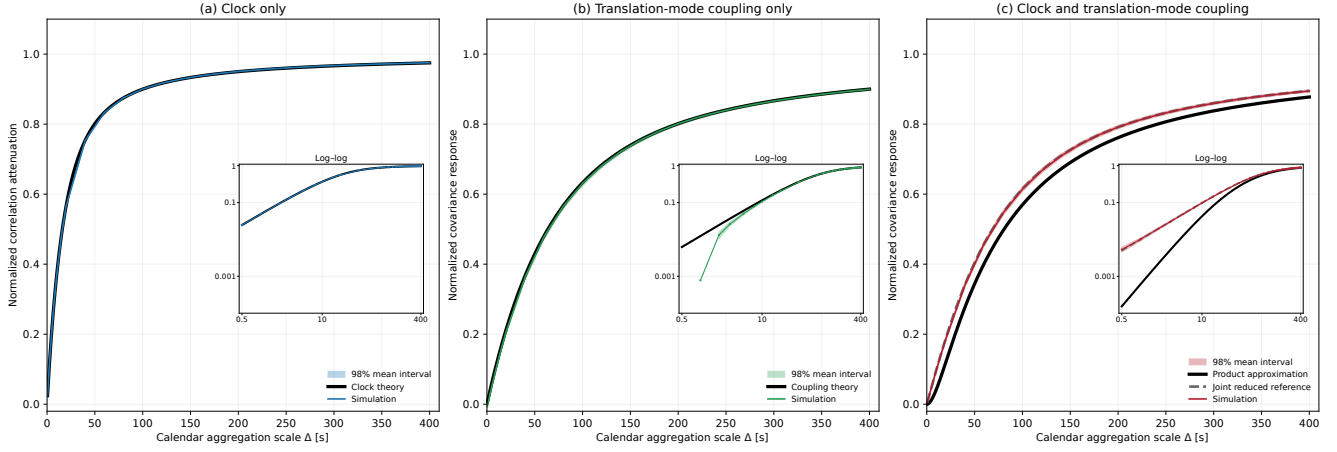


Figure 7: Clock and translation-mode Epps comparisons. Three square panels share aggregation scales and vertical limits. (a) Blue: matched clock attenuation; black: equal-rate clock factor, with each book's rate 0.1 s^{-1} . (b) Green: matched coupling covariance; black: coupling factor with $\kappa = 0.025 \text{ s}^{-1}$. (c) Red: matched combined covariance; black: product approximation; dashed grey: stationary joint reduced reference $G(\Delta)$ in Eq. (30). Shading denotes approximate pointwise 98% mean intervals across 512 independent paths, grouping four clock replicates per path where applicable. Maximum half-widths are 0.00217, 0.00749 and 0.00717 respectively. The narrow band in (c) resolves the product-approximation difference; it is not an individual-path envelope. No parameters are refitted. Insets repeat the curves and intervals on log-log axes over the measured 0.5–400 s range. The smallest-lag coupling-only points remain sensitive to the update size; nonpositive values are shown only on the linear axes. Separate square exports accompany this combined figure.

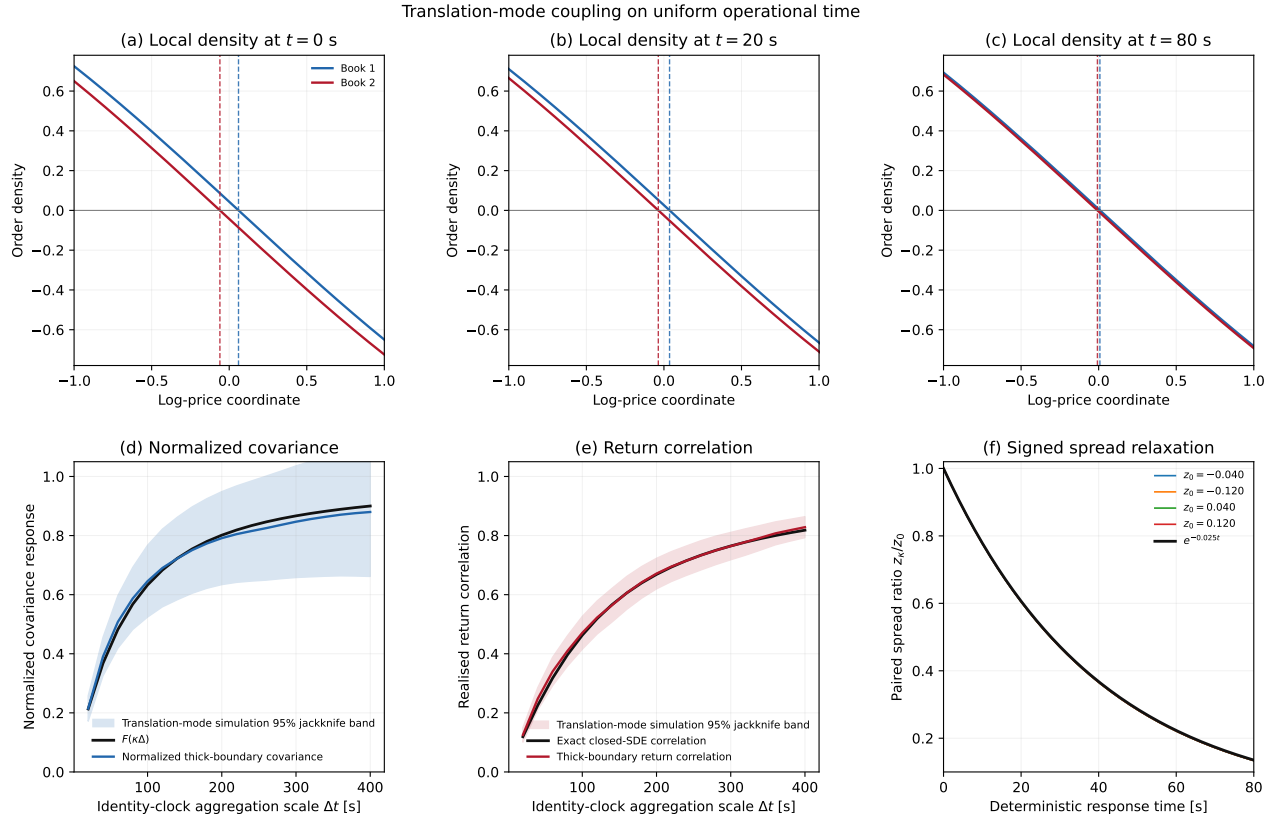


Figure 8: **Translation-mode coupling on uniform operational time.** Retained v2.1.0 component diagnostic. Holdout thick-boundary paths are compared with the analytical normalized covariance response and the distinct finite-scale return correlation. The covariance rate is estimated on a disjoint calibration ensemble, as in the published diagnostic. Lower panels verify signed deterministic front relaxation and residuals at the frozen response rate. No clock, interpolation or parameter fit enters this comparison.

Paired single-trade own and cross impact

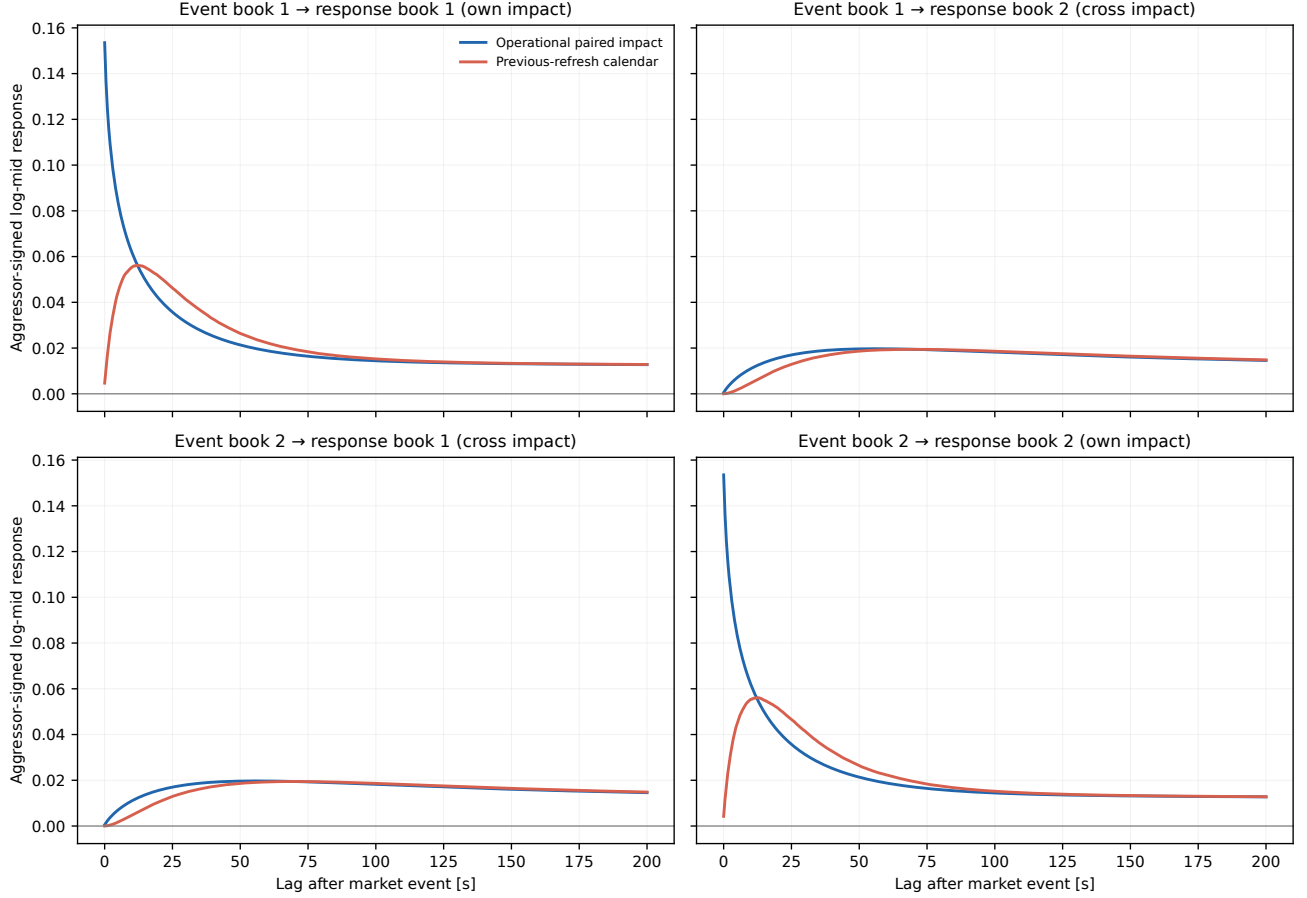


Figure 9: **Paired single-trade own and cross impact.** Rows identify the event book and columns the response book, preserving the full directional matrix on a common linear scale. Blue denotes operational impact; red denotes Poisson previous-refresh impact. Curves average 512 independent primitive groups, with signs, four symmetry partners and four clock replicates grouped before calculating pointwise 95% mean intervals. The order quantity is 0.05, $\Delta x = 0.05$, and the calendar-equivalent update is 0.125 seconds. Physical injection occurs at 150 seconds; displayed lag zero retains the recorded 150.5-second completion anchor. Clock-inactive observations are included. Bands describe sampling uncertainty and do not include spatial discretization error.

Meta-order impact: equal signed volume, distinct execution horizons, explicit subordination

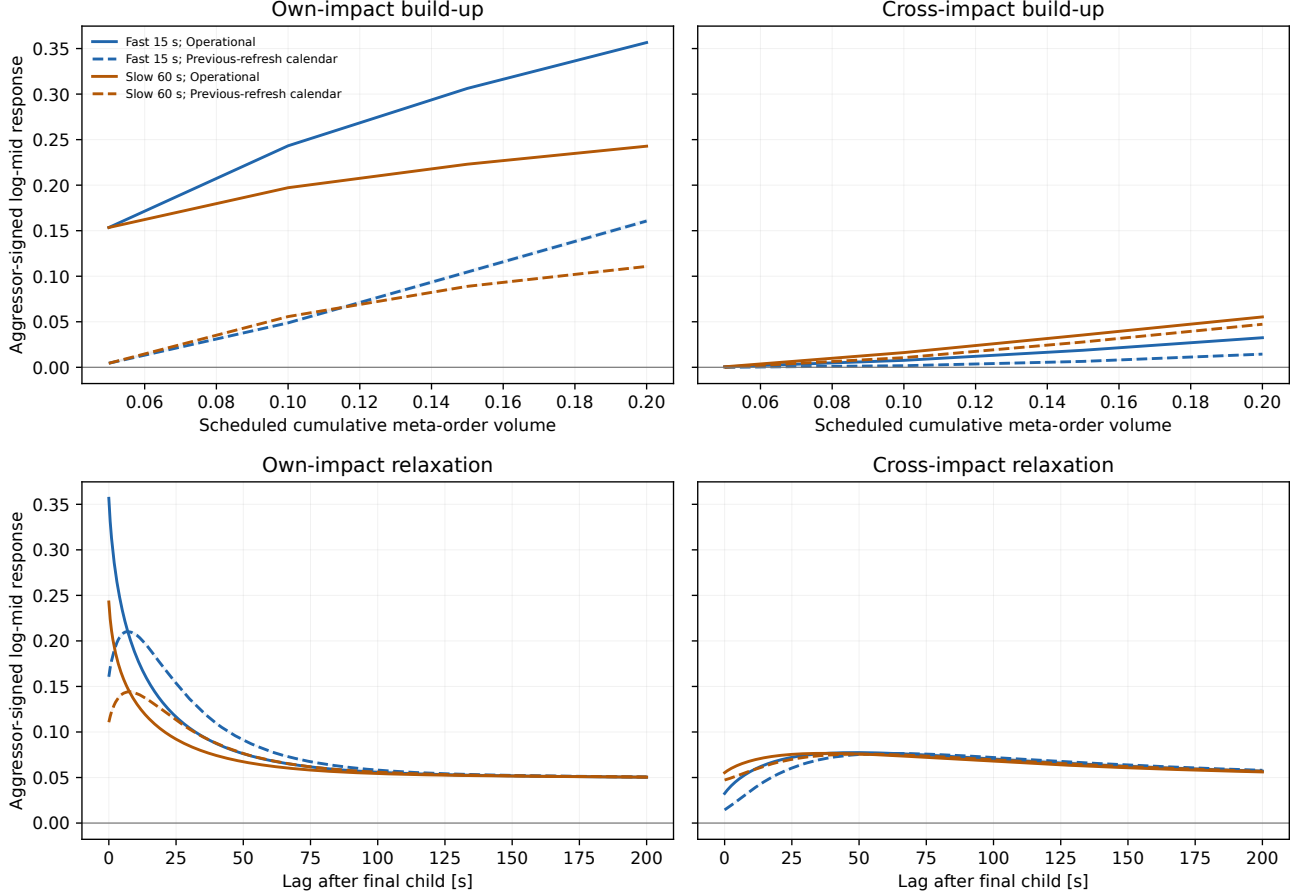


Figure 10: **Scheduled meta-order own and cross impact.** Build-up is above relaxation; own and cross responses occupy the two columns on a shared vertical scale. Blue denotes the fast 15-second schedule and brown the slow 60-second schedule; solid denotes operational observation and dashed Poisson previous refresh. Four children of quantity 0.05 give total signed volume 0.20. Means and pointwise 95% intervals use 512 independent groups with the grouping and $\Delta x = 0.05$ grid of Figure 9. Physical injection times are preserved under refinement. Relaxation lag zero is the recorded completion anchor, 0.5 seconds after the final physical injection. Equal volume and distinct execution duration are schedule controls, not estimates of a market participation rate.

Mid-price and trade-sign dependence: event time, operational time and explicit subordination

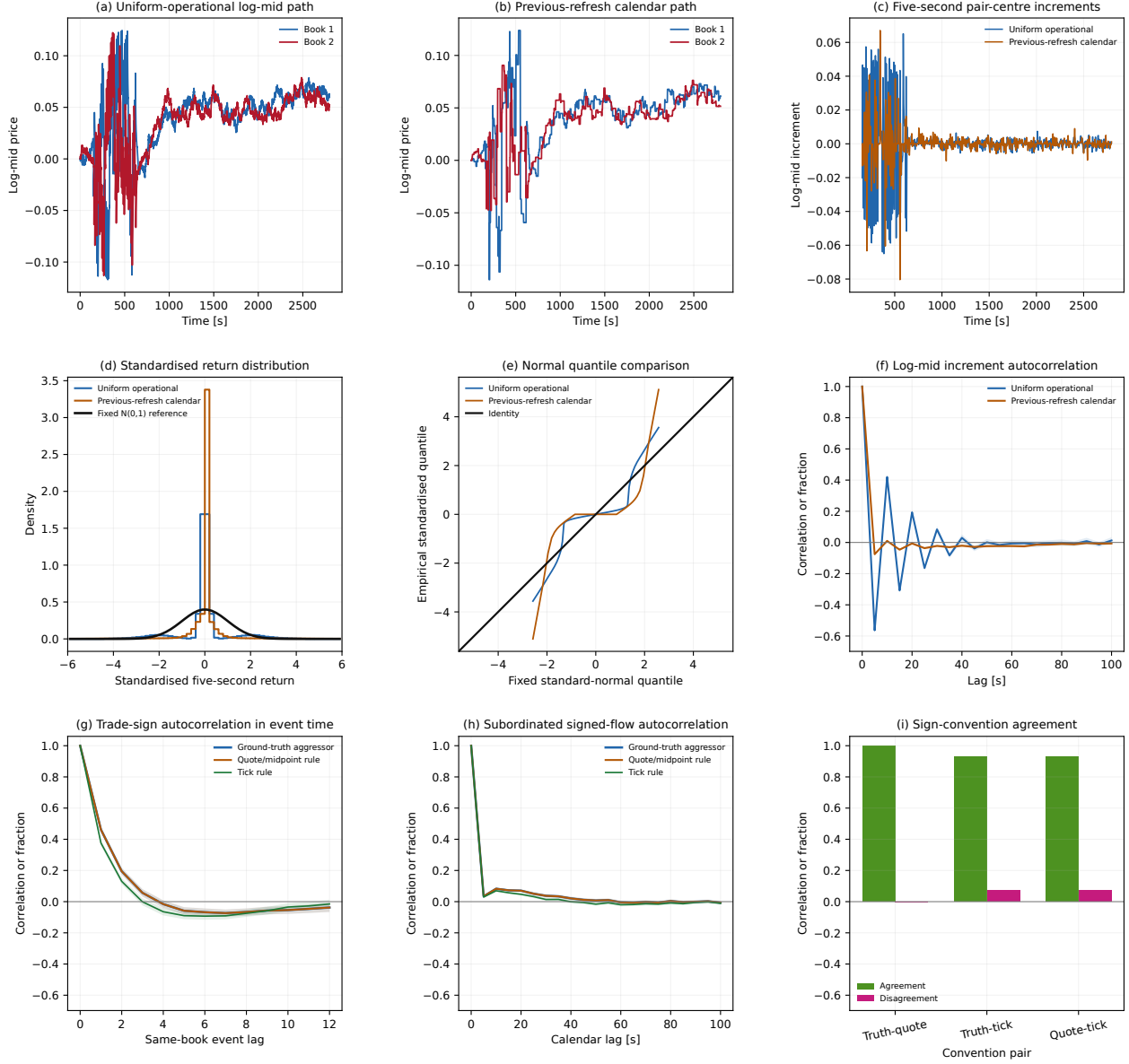


Figure 11: **Mid-price and trade-sign dependence.** The original nine-panel layout separates example paths and pair-centre increments (a–c), pooled standardized returns and normal quantiles (d–e), return autocorrelation (f), event-time sign autocorrelation (g), subordinated flow autocorrelation (h), and sign agreement (i). The example is the predeclared primitive group 2. Ensemble summaries use 128 independent primitive groups with explicit antithetic partners, 5600 half-second steps, $\Delta x = 0.1$, and five-second returns after step 301. Each book has 48 quantity-0.015 events at the same fixed schedule; sign repeat probability is 0.75. Bands are pointwise 95% intervals across independent groups. Extending the horizon adds post-execution time, so distributions and ACFs are horizon-dependent; periodic-schedule structure is not removed by smoothing. Normal curves are references, not fitted targets. These are finite-schedule diagnostics, not an endogenous long-memory claim.

where H is the transport or memory contribution,

$$A_{j,n} = \Delta u q_j, \quad C_{j,n} = \left(e^{-\nu_j \Delta u} - 1\right) \varphi_{j,n}^+, \quad L_{j,n} = \Delta u \ell_{j \leftarrow k}, \quad (36)$$

and B is the imposed outer-boundary correction. The plotted arrivals, removals and market-order impulse are density contributions rather than rates. All are drawn on the same density scale. In particular, the green curve is $A_{j,n} = \Delta u q_j$ and is small relative to φ_j because $\Delta u = 0.005$; it is not independently rescaled. The unplotted terms are retained in the machine-readable ledger.

Two differences from Bauer et al. Figure 2 follow from the present model. The accepted configuration has $\nu_1 = \nu_2 = 0$, so $C_{j,n} = 0$ and the gold removal curve is identically zero. It is retained to preserve the plotted object map; no positive cancellation rate is introduced for visual agreement. Second, boundary-outward consumption produces a short zero-density interval in the intermediate 0^+ state. That state has no unique simple zero crossing. The marker in panel (b) is therefore the last registered boundary p_1^- ; the first completed fixed-grid step restores a unique registered reaction boundary. Later markers are independently recovered from the corresponding density states.

The first completed step gives an own-boundary displacement of 0.0997 in log-price units. The displacement reaches 0.111 at 2 seconds and then decreases to 0.0160 at 80 seconds. Thus the computed late state is substantially relaxed but remains displaced from the matched control over the displayed horizon.

The calculation uses fixed $\Delta x = 0.1$ and $\Delta u = 0.005$ model-time units, with 100 seconds per model-time unit. Innovations are set to zero so that the chronology isolates event consumption, diffusion, source renewal and coupling. A matched no-event control remains in the numerical checks but is not added to the nine-panel figure. The result recovers the structure and scientific question of the earlier figure; it does not assert numerical identity with its non-uniform-time trajectory or parameterisation.

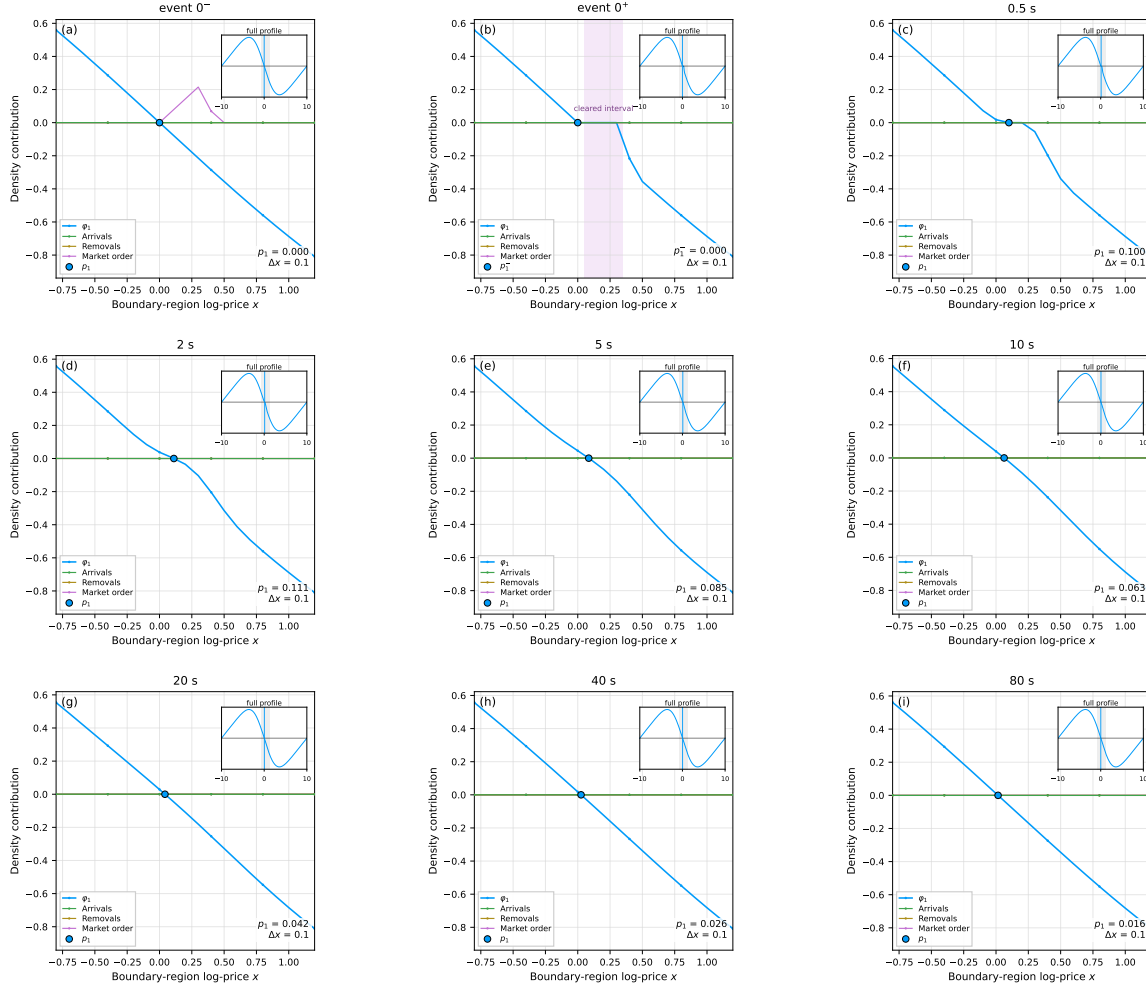


Figure 12: **Fixed-time order-book shock recovery.** Nine snapshots follow a buy market order in book 1 from the pre-event state 0^- , through the post-consumption state 0^+ , to 80 seconds of fixed operational-time evolution. Blue is the signed density φ_1 , green is the per-step arrival contribution, gold is the per-step removal contribution, magenta is the market-order density increment shown in panel (a), and the blue marker is the registered reaction boundary. The order consumes opposing liquidity from the boundary outwards. In panel (b) this creates a zero-density interval, so the marker is the last registered boundary p_1^- ; a unique simple zero is again registered after the first completed step. The accepted model has $\nu_1 = \nu_2 = 0$, and the gold curve is consequently zero in every panel. Translation-mode coupling to book 2 remains active and is retained in the numerical ledger but is not drawn as an additional curve. The registered boundary moves by 0.0997 after the first completed step, reaches 0.111 at 2 seconds and relaxes to a residual displacement of 0.0160 at 80 seconds. The price grid and operational step are fixed. All contribution curves use the same density scale and are not independently rescaled. The figure recovers the earlier panel structure without claiming a pointwise replication of the Bauer et al. trajectory.

9.2 Long-memory order flow and observation-clock morphology

Figure 13 separates two mechanisms that can otherwise be confused. The operational solver produces one completed pair of price paths on the fixed grid. Its innovations are Gaussian, while the declared market-order signs are an exogenous order-splitting input. Independent book-specific renewal clocks are then applied by the exact previous-refresh map in Eq. (17). Thus “Gaussian” labels the operational innovations and never a waiting-time distribution.

For each primitive path and book, signs are constant over iid meta-order run lengths L with tail exponent $\nu = 1.5$ and minimum run length two events; successive run signs are iid and symmetric. In the corresponding ideal renewal construction the sign covariance decays as a power law proportional to $k^{1-\nu}$. This is a declared long-memory input motivated by order splitting, not an endogenous consequence of the order-book dynamics and not an empirical calibration.

The four rows hold the operational dynamics, event tape and query grid fixed. They differ only in observation:

1. the completed operational path itself;
2. exponential waiting intervals with rate 0.1 s^{-1} ;
3. Mittag–Leffler waiting intervals with $\beta = 0.8$ and $\tau = 10 \text{ s}$; and
4. the exact exponential tilt of that Mittag–Leffler law with $\lambda_T = 0.0125 \text{ s}^{-1}$.

The resulting zero-return proportions at the five-second query scale are, respectively, 0, 0.6063, 0.9097 and 0.6648. The central spike, QQ plateau and apparent tail thickening therefore measure observation-clock morphology. Untempered subdiffusion produces the longest holds; tempering reduces, but does not remove, that effect. A constant finite-horizon sampled series is retained in distribution and impact averages, but its undefined ACF is not assigned an artificial value. Group ACFs require both books and both antithetic partners to be defined. At the longest horizon, the Mittag–Leffler row has between 121 and 125 of 128 valid groups, depending on lag and observable; this conditional availability is distinct from the unconditional return distribution.

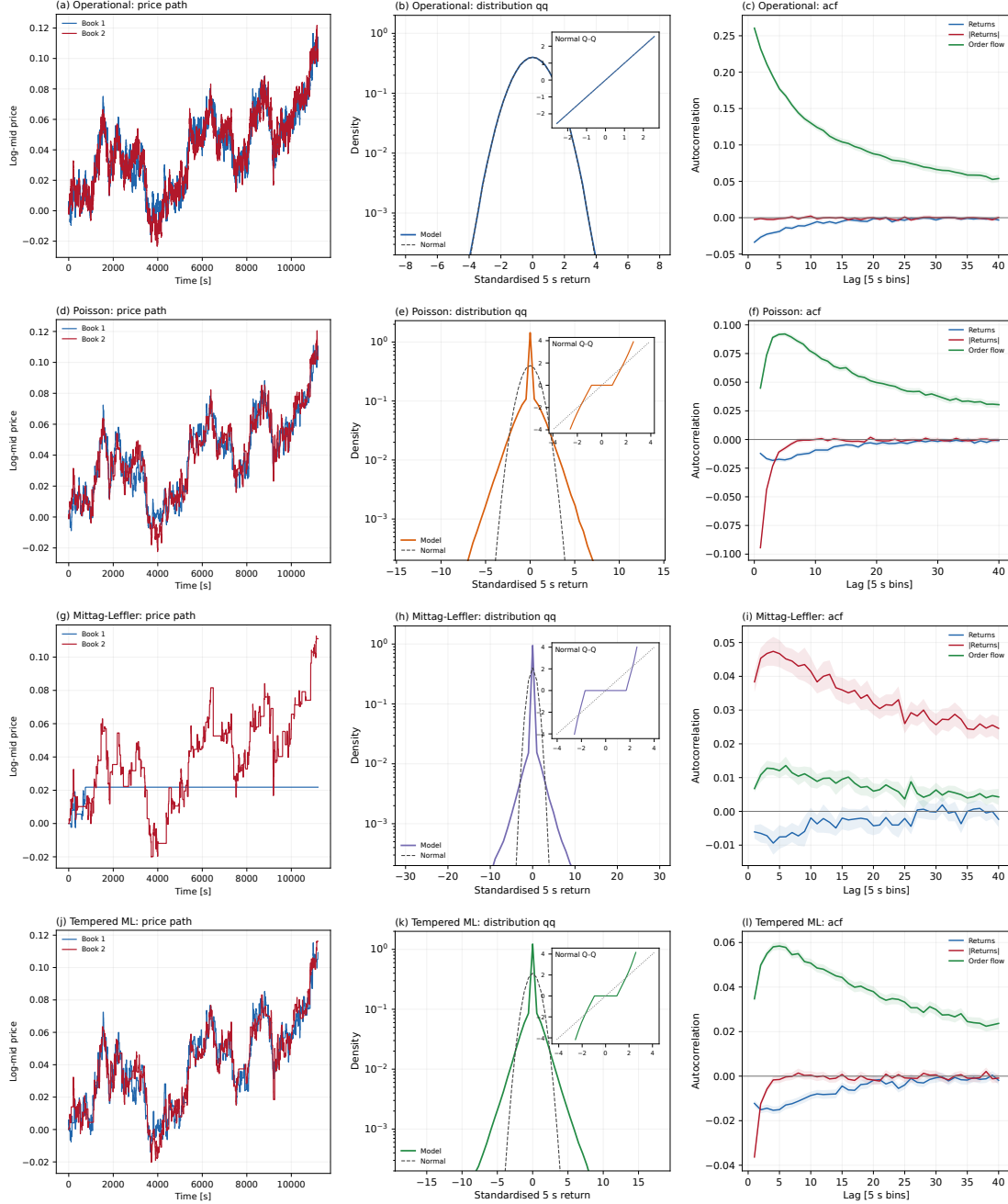


Figure 13: **Operational long-memory input and observation-clock morphology.** The four rows show operational, Poisson, Mittag–Leffler and tempered Mittag–Leffler observation. The left column is the predeclared primitive group-2 path, with blue/red denoting books 1/2. The centre column shows pooled standardized five-second return densities on a logarithmic scale, with a fixed normal reference and QQ insets. The right column shows return (blue), absolute-return (red) and declared-flow (green) ACFs at lags in five-second bins; shading is pointwise 95% mean uncertainty. Summaries use 128 independent groups with explicit antithetic partners, 22400 half-second steps and $\Delta x = 0.1$. Exogenous quantity-0.0002 events arrive with probability 0.05 per step and Pareto sign-run exponent 1.5; only the longest horizon has the terminal 149-step no-event margin. Clocks start at time zero. Infinite-mean renewal aging makes observation-window length material. Constant paths remain in densities; undefined ACFs are omitted only at the declared complete-group level. Gaussian innovations do not require Gaussian observed returns, and no tail or clock parameters are refitted.

9.3 Mittag–Leffler renewal clocks

Let $W_{\beta,\tau}$ denote a positive Mittag–Leffler waiting interval. The implemented convention is fixed by its Laplace transform,

$$\mathbb{E}[e^{-sW_{\beta,\tau}}] = \frac{1}{1 + (\tau s)^\beta}, \quad 0 < \beta \leq 1, \quad \tau > 0. \quad (37)$$

For $0 < \beta < 1$ it has infinite mean. If E is unit exponential and S_β is positive stable with $\mathbb{E}[e^{-sS_\beta}] = e^{-s^\beta}$, the sampler uses $W_{\beta,\tau} = \tau E^{1/\beta} S_\beta$. The positive-stable draw uses the Kanter representation in Algorithm 3.

Exponential tempering is defined as a probability-law tilt, not by truncating realised waits. For rate $\lambda_T > 0$ its transform and mean are

$$\mathbb{E}[e^{-sW_{\beta,\tau}^{(T)}}] = \frac{1 + (\tau\lambda_T)^\beta}{1 + [\tau(s + \lambda_T)]^\beta}, \quad (38)$$

$$\mathbb{E}[W_{\beta,\tau}^{(T)}] = \frac{\beta\tau^\beta\lambda_T^{\beta-1}}{1 + (\tau\lambda_T)^\beta}. \quad (39)$$

Exact rejection sampling accepts an untempered candidate w with probability $e^{-\lambda_T w}$. The retained v2.1.0 law controls over 240,000 supplied candidate uniform quadruples give maximum Laplace-transform errors 9.76×10^{-4} and 1.03×10^{-3} for the untempered and tempered laws, respectively; the tempered mean has relative error 2.67×10^{-3} . The 99th-percentile wait falls from 487.6 s to 78.2 s under tempering. Random-number streams remain caller-owned and book-specific.

9.4 Paired price and meta-order impact across clocks

The implementation audit used Diana and Gebbie’s [InteractingLOBs.jl](#) at commit 098f1807. Its substantive routine `src/useful_functions.jl:calculate_price_impacts` measures signed raw-price displacement after an injected order and aggregates paths; the repository-root `plot_price_impact.jl` is only a stub. The associated method is described in [Diana and Gebbie, *J. Comput. Appl. Math.* 2024](#) and arXiv:2310.06079. The Julia routine is a methodological reference and is not ported. The present implementation uses the common-input shocked-minus-control estimator established in Figures 9 and 10.

For event book b , response book j , sign $s \in \{-1, +1\}$, path p and clock domain c , the reported single-trade estimand is

$$\hat{I}_{b \rightarrow j}^{(c)}(\ell) = \frac{1}{2P} \sum_{p=1}^P \sum_{s \in \{-1, +1\}} s \left[P_{p,j}^{(c),\text{shock}}(t_e + \ell) - P_{p,j}^{(c),\text{control}}(t_e + \ell) \right]. \quad (40)$$

The shocked and control paths use identical initial states and operational innovations. For each observation domain they also use the same realised book clock, so a response can be delayed or held but not generated by a clock mismatch. The meta-order estimand applies Eq. (40) to the declared four-child fast and slow schedules and reports lag from the last child. Active-observation fractions are stored with the curves; inactive paths contribute zero at that calendar query, preserving the unconditional clock effect.

Paired price impact under alternative observation clocks

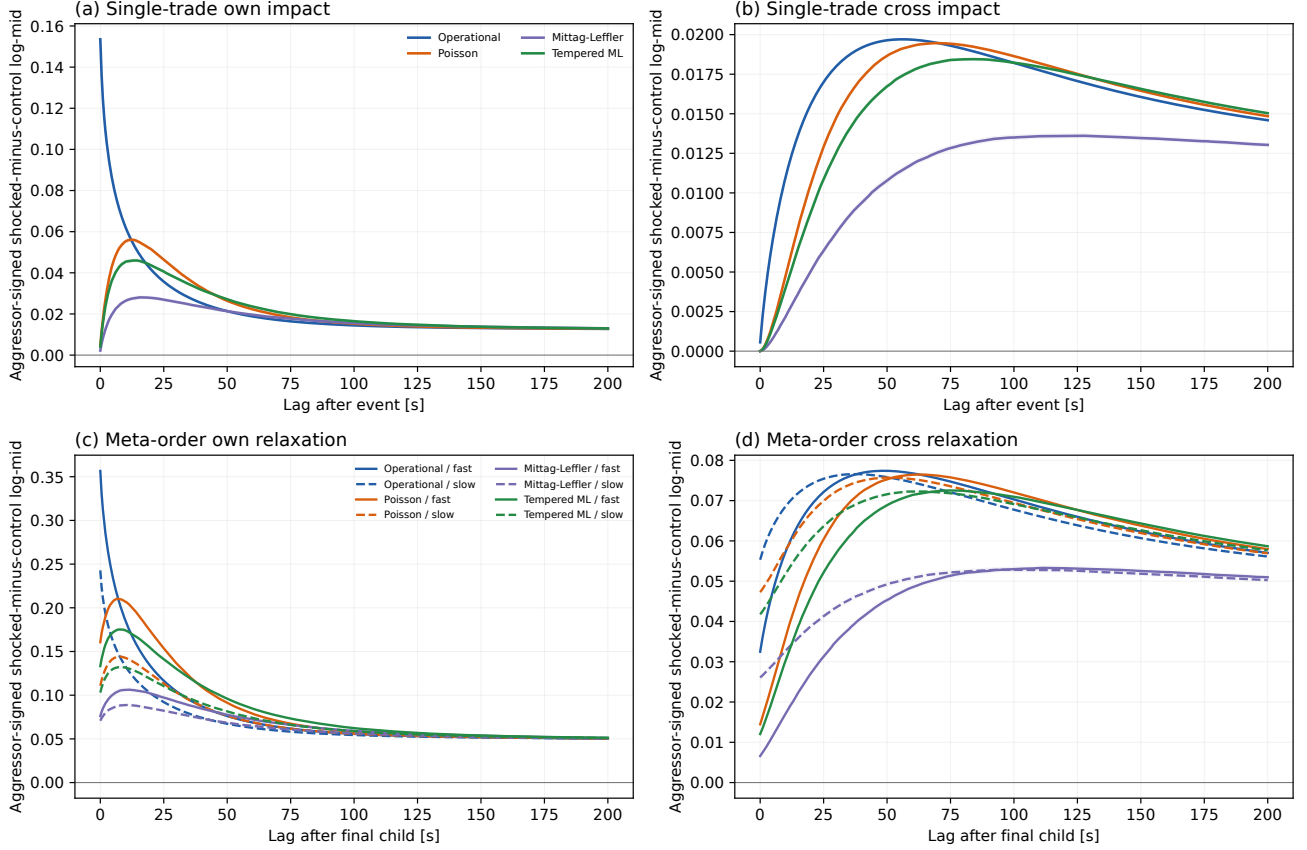


Figure 14: **Paired price impact under alternative observation clocks.** Single-trade own/cross impact is above fast/slow meta-order relaxation. Blue, orange, purple and green denote operational, Poisson, Mittag–Leffler and tempered Mittag–Leffler observation. Fast schedules are solid; slow schedules are dashed. Means and pointwise 95% intervals use 512 independent primitive groups, with symmetry partners and four clock replicates grouped within each primitive. The grid is $\Delta x = 0.05$ with a 0.125-second update; physical event times and the recorded completion anchors match Figures 9 and 10. Clock-inactive observations remain in the average. Narrow bands measure mean uncertainty, not individual-response dispersion. Broad 10th–90th percentile individual meta-order envelopes are supplied as a [separate companion figure](#). Heavy-tailed waits can delay observation throughout this finite horizon; no universal long-lag convergence is asserted for the infinite-mean clock. These are conditional model responses, not empirical impact laws.

Table 1: **Parameter, timescale and identifiability register.** The table maps paper notation to implementation names and records units, roles, sensitivity pathways, and fixed or swept status. Refresh rate and source amplitude receive different code names despite source-notation overlap. Identifiability classifications describe the information available from the planned outputs under the stated model; they are not estimates, formal statistical-identifiability proofs, or empirical calibration results.

Paper bol	sym-	Code name	Source	Units	Role	Sensitivity path	Figure treatment	Identifiability from Epps output
Δ		<code>aggregation_scale</code>	Eqs. (8), (9), (11)	time; dimensionless after scaling	return aggregation horizon	direct horizontal scale	swept in F1-F6	observed/design variable
ρ_∞		<code>rho_inf</code>	Eqs. (8), (9), (11)	dimensionless correlation	long-scale vertical level	multiplicative vertical scale	fixed to one in base figures	conditional on plateau coverage
λ_{12}		<code>clock_rate</code>	Eq. (8); App. A	time ⁻¹	pooled re- fresh/overlap rate	$F(\lambda_{12}\Delta)$	unit scale; sensitivity in F2	conditional; independent- ly measurable in v2
$\kappa = \kappa_1 + \kappa_2$		<code>response_rate</code>	Eq. (9); App. B	time ⁻¹	spread relaxation rate	$F(\kappa\Delta)$	swept in F1-F2; de- rived in F4-F5	conditional on clock rate and short scales
α_c		<code>alpha_clock</code>	App. C, Eq. (C.18)	dimensionless	fractional clock order	short power Δ^{α_c}	swept in F3	not separate from <code>alpha_response</code> at short scale
α_r		<code>alpha_response</code>	App. C, Eq. (C.18)	dimensionless	fractional response order	short power Δ^{α_r}	swept in F3	not separate from <code>alpha_clock</code> at short scale
τ_c, τ_r		<code>clock_characteristic_time</code> <code>response_characteristic_time</code>	v1 nondimension- alisation	time	fractional scale con- vention	$(\Delta/\tau)^\alpha$	fixed to one in F3	confounded with frac- tional rate convention
γ_{jk}		<code>coupling_strength</code>	Eq. (6); App. B	model-dependent	pair-trader coupling strength	$\kappa_j \propto \gamma_{jk}$	one-at-a-time sweep in F4	not separate from source/front quanti- ties
λ_j (source)		<code>source_amplitude</code>	Eq. (5); App. B	source amplitude	decaying-Gaussian amplitude	$M_j^+ \propto a_j$	one-at-a-time sweep in F4	not separate from coupling strength
μ_j		<code>source_width</code>	Eq. (5); App. B	price ⁻²	Gaussian source- shape scale	$M_j^+ \propto \mu_j^{-1/2}$	one-at-a-time sweep in F4	not separate from am- plitude/front slope
$ \mathcal{L}_j $		<code>front_slope_abs</code>	Eq. (B.9); Eq. (B.13)	density per price	frozen reaction-front slope	$\kappa_j \propto \mathcal{L}_j ^{-1}$	one-at-a-time sweep in F4	not identifiable as thickness from Epps alone
ε		<code>epsilon</code>	Eq. (6); App. D	price ² for yz/ε	continuum selector regularisation	centred moment in- variant; representa- tion path	swept in F5	not a reaction-front thickness parameter
Δx		<code>lattice_spacing</code>	App. D	price	numerical lattice res- olution	discrete moment er- ror to effective re- sponse	swept in F5	known numerical con- trol
z_{jk}		<code>directed_spread</code>	Eq. (6); App. B	price	directed book-price spread	selector argument yz/ε	fixed to 0.2 in F5	state variable, not in- ferred parameter
–		<code>alignment_offset_cells</code>	v1 numerical diag- nostic	lattice cells	source-selector cen- tring error	parity breaking to ef- fective response	0 and 0.5 in F5	known numerical con- trol
–		<code>domain_halfwidth</code>	v1 numerical diag- nostic	price	finite integra- tion/lattice domain	tail truncation to ef- fective response	full and truncated in F5	known numerical con- trol

Table 2: **Numerical benchmarks, convergence tests and acceptance status.** Each row records a claim-relevant diagnostic, its analytic or independent reference result, declared tolerance, observed value or error, and acceptance status. Checks cover the ordinary kernel and derivative, fractional $\alpha = 1$ recovery and independent reference values, the decaying-Gaussian first moment, centred-selector invariance, finite-grid refinement, invalid inputs, and the analytic/simulation overlay contract. The table supports numerical reliability and reproducible execution within the tested environment; it is not a mathematical proof of the model, an empirical validation, or evidence for an untested simulation model.

ID	Diagnostic	Expected result	Tolerance	Observed result	Status
D01	Ordinary zero and small-argument limit	$F(0)=0$ and the declared small-x series is recovered	abs error $\leq 1.0\text{e-}12$	$F(0)=0.000\text{e}+00$; series error at $1\text{e-}6=0.000\text{e}+00$	Verified
D02	Ordinary bounds and monotonicity	$0 \leq F(x) < 1$ and non-decreasing for $x \geq 0$	violation $\leq 1.0\text{e-}12$	range=[0.00000500, 0.99000000]; min increment= $1.641\text{e-}07$	Verified
D03	Ordinary analytic derivative	Analytic derivative agrees with centred finite differences	max error $\leq 2.0\text{e-}07$	max error= $2.084\text{e-}10$	Verified
D04	Ordinary elasticity limits	$S_F \rightarrow 1$ at short scale and $S_F \rightarrow 0$ at long scale	combined limit error $\leq 2.0\text{e-}05$	$S(1\text{e-}8)=0.999999996667$; $S(1\text{e}5)=1.000\text{e-}05$	Verified
D05	Fractional alpha-one recovery	F_α with $\alpha=1$ equals the ordinary kernel	max error $\leq 2.0\text{e-}13$	max error= $0.000\text{e}+00$	Verified
D06	Fractional independent reference values	Release evaluator matches adaptive-quadrature references	max error $\leq 2.0\text{e-}09$	six-point max error= $1.571\text{e-}13$	Verified
D07	Fractional bounds and monotonicity	Release curves remain in $[0,1]$ and non-decreasing	violation $\leq 2.0\text{e-}09$	maximum violation= $0.000\text{e}+00$	Verified
D08	Decaying-Gaussian first moment	Numerical full-domain selected moment equals analytic half-line moment	relative error $\leq 2.0\text{e-}06$	analytic= -0.443113462726 ; numerical= -0.443113462726	Verified
D09	Centred-selector continuum invariance	Symmetric first moment is independent of selector width	relative error $\leq 2.0\text{e-}06$	max relative error over three epsilon values= $2.220\text{e-}16$	Verified
D10	Centred finite-grid convergence	Moment error decreases from a deliberately coarse lattice under refinement	finest error $\leq 2.0\text{e-}05$	errors= $1.935\text{e-}01, 1.939\text{e-}03, 1.635\text{e-}06, 1.110\text{e-}15$	Verified
D11	Boundary baseline response normalisation	Two-book baseline total response rate equals one	abs error $\leq 1.0\text{e-}12$	$\text{kappa_total}=1.000000000000$	Verified
D12	Combined-factor construction	Combined curve equals diagnosed component product and remains bounded	violation $\leq 1.0\text{e-}12$	range=[$0.00000025, 0.98010000$]	Verified
D13	Invalid-input handling	Four invalid parameter cases fail clearly	all four raise ValueError	caught=4/4	Verified
D14	v2 analytic/simulation overlay contract	Required scientific and provenance roles are declared	all ten roles present	declared roles=10/10	Verified

10 Parameter and numerical evidence

Table 1 keeps the source notation, unambiguous code names, units, sensitivity paths, and identifiability qualifications together. In particular, the paper’s refresh-rate and source-amplitude uses of λ are separated in code. Table 2 retains the published deterministic diagnostic results and tolerances; it does not certify the new stochastic precision or continuum limit. The fractional reference points were computed independently with adaptive quadrature, while the release evaluator uses fixed composite Gauss–Legendre quadrature and a small-argument series.

11 Numerical resolution and reproducibility

Comparison	Independent groups	Resolution and uncertainty
Clock and coupling (7a–c)	512 validation	8000 s after 200 s warm-up; $\Delta x = 0.1, 0.5$ s update; 98% pointwise ratio intervals.
Impact (9, 10, 14)	512 primitive	$\Delta x = 0.05, 0.125$ s update; 161 lag points; 95% pointwise mean intervals.
Finite-schedule dependence (11)	128 primitive	5600 half-second steps; explicit antithetic partners; 95% pointwise intervals.
Persistent-flow morphology (13)	128 primitive	22400 half-second steps; explicit antithetic partners; 95% pointwise intervals where defined.

The short-lag grid comparison is discussed with Eq. (32); it does not change the following precision statements for the previously sampled ranges. At the original 95% precision criterion, the clock-only maximum half-width is 0.00182. Coupling-only and combined half-widths are 0.00631 and 0.00604, exceeding the proposed 0.002 target. Nineteen of 24 impact schedule–clock–response combinations meet the fixed plot-scale precision criterion; five, primarily involving renewal-clock cross impact, remain above it. These results are retained without changing seeds or fitting parameters. The finer $\Delta x = 0.025$ impact screen still changes early own impact by approximately 0.001–0.003; the production $\Delta x = 0.05$ grid is not a certified continuum limit. The 128-group persistent-flow study meets its longest-horizon positive-lag ACF precision targets, but finite-schedule and horizon effects remain.

The source boundary is the published v2.1.0 commit `e932f7c66449d4fd5e0269cbcaafa324d91b51bf`. The numerical figures are regenerated from the declared settings and fixed seeds through the existing entry point:

```
python scripts/run_all.py
```

To compile this supplement from the repository root, run the following command twice:

```
pdflatex -output-directory=supplementary-materials SUPPLEMENTARY-MATERIAL-v2.2.0.tex
```

Document compilation uses the generated figures; it does not rerun simulations.

The retained figures are verified against their immutable supplied assets. The full v2.1.0 release remains available independently. PDF bytes may differ across renderers, fonts and platforms; machine-readable values, seed rules, input hashes, tolerances and declared statistical units are the scientific reproducibility objects. The source asset and its accepted data are preserved.

12 Interpretive limits

The diagnosed computations support a limited set of conclusions. Ordinary clock and coupling factors are distinguishable when their timescales are sufficiently separated and the components are shown explicitly. The long-scale plateau contains little local information about either rate. Within the stated fractional product ansatz, the leading short-scale power identifies an order sum rather than two separate orders; this is not a general result for a joint clock–coupling process. Reaction-boundary quantities affect the Epps curve conditionally through the effective response rate, but are not individually identified by that curve. Continuum selector invariance can coexist with finite-grid distortion when parity, alignment or domain controls are not respected. The clock-only matched statistic is close to its theoretical factor at the plotted scale. Coupling comparisons retain short-scale residuals, and the combined product is a separability approximation with a different short-time power from the joint reduced process. Reduced conditional moments account for realized clock indices without proving identity with the full reaction–diffusion process. Heavy-tailed sign runs establish a controlled long-memory input but do not show that the book endogenously generates that memory. Mittag–Leffler clocks change observed return morphology and delay observed impact without changing the underlying operational response. The impact, shock-recovery, autocorrelation and stylised-facts figures are conditional model diagnostics, not empirical calibration.

These are computational illustrations and numerical verification statements inside the paper’s assumptions. They are not empirical validation, calibration or proof of the approximations. The antecedent implementation is not required to execute or interpret these comparisons.

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